

EEG-Based Epileptic Seizure Prediction Using Machine Learning

Comparative Analysis of Preprocessing Pipelines, Regularization
Techniques, and Class Imbalance Handling

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Semester Project Report

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Abstract

This project presents a comparative machine learning study for EEG-based epileptic seizure prediction using Logistic Regression under multiple preprocessing and regularization strategies. The study investigates the impact of feature standardization, class imbalance handling using SMOTE, and regularization techniques including L1, L2, and Elastic Net regularization.

Two EEG datasets were used for experimentation: the Federated EEG dataset and the Epileptic Seizure Recognition dataset. Exploratory Data Analysis (EDA), confusion matrix analysis, ROC analysis, Precision-Recall evaluation, and class distribution analysis were performed to evaluate model behavior.

Experimental findings demonstrate that severe class imbalance significantly affects seizure prediction performance. Although baseline Logistic Regression achieved higher overall accuracy, it failed to predict minority seizure classes effectively. After applying SMOTE, minority class learning improved considerably, resulting in more balanced classification performance.

The project highlights the importance of preprocessing pipelines, regularization strategies, and balanced learning approaches in EEG-based seizure detection systems.

1 Introduction

Epilepsy is a neurological disorder characterized by abnormal electrical activity in the brain that may lead to repeated seizures. Accurate seizure prediction can help clinicians monitor patients, improve diagnosis, and reduce medical risks.

Machine learning techniques have become increasingly important in biomedical signal processing because they can automatically identify hidden patterns within EEG signals. However, EEG datasets often contain high-dimensional features, noisy measurements, and severe class imbalance, which make seizure prediction challenging.

This project focuses on evaluating Logistic Regression-based seizure prediction under different preprocessing and regularization strategies.

2 Project Objectives

- To develop a machine learning pipeline for EEG seizure prediction
- To compare preprocessing techniques for EEG data
- To evaluate Logistic Regression under multiple regularization methods
- To investigate the impact of SMOTE on class imbalance
- To analyze confusion matrices, ROC curves, and Precision-Recall curves
- To evaluate cross-dataset generalization performance

3 Datasets Used

Table 1: Datasets Used in the Project

Dataset	Purpose
Federated EEG Dataset	Multi-class EEG seizure classification and imbalance analysis
Epileptic Seizure Recognition Dataset	Cross-dataset validation and seizure prediction evaluation

4 Environment and Tools

The experiments were conducted using Python in Jupyter Notebook.

4.1 Libraries Used

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Imbalanced-learn (SMOTE)

5 Dataset Exploration and EDA

Initial exploratory analysis was performed to understand dataset structure, feature composition, class distributions, and data quality.



Figure 1: Class Distribution of EEG Dataset

The Federated EEG dataset showed noticeable class imbalance where the majority class dominated minority seizure classes.

6 Dataset Cleaning and Preparation

The datasets were prepared for machine learning experiments by:

- Separating features and target labels
- Handling missing values
- Removing unnecessary columns
- Standardizing EEG features

Feature standardization was performed using StandardScaler to improve model convergence and regularization performance.

7 Baseline Logistic Regression Model

A baseline Logistic Regression model was trained using standardized EEG features.

The baseline model achieved moderate overall accuracy; however, the classification report revealed strong majority-class bias.

8 Confusion Matrix Analysis

Confusion matrices were generated to evaluate model predictions across seizure classes.

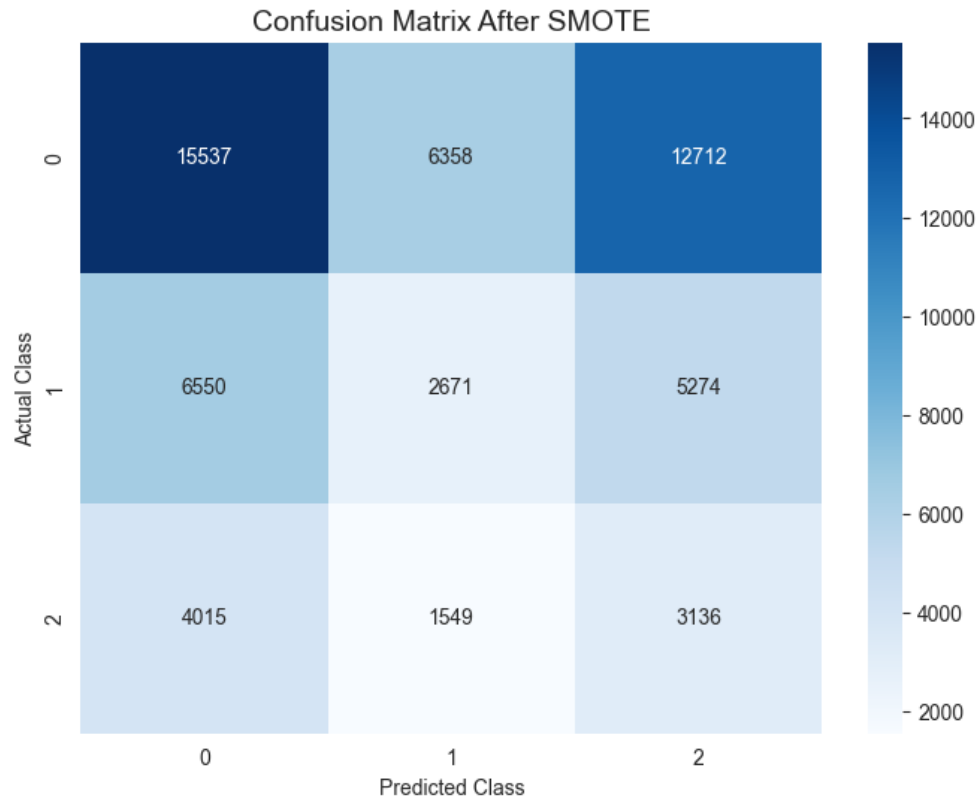


Figure 2: Confusion Matrix for Logistic Regression

The confusion matrix analysis demonstrated that baseline Logistic Regression heavily favored the majority class while SMOTE-based models improved minority class predictions.

9 SMOTE-Based Class Imbalance Handling

Synthetic Minority Oversampling Technique (SMOTE) was applied to address severe class imbalance.

After applying SMOTE:

- Dataset class distribution became balanced
- Minority class recall improved
- Model predictions became more stable
- Overall accuracy slightly decreased due to balanced learning

Although the accuracy reduced slightly, seizure prediction quality improved significantly.

10 ROC Curve Analysis

Receiver Operating Characteristic (ROC) analysis was conducted to evaluate classification capability.

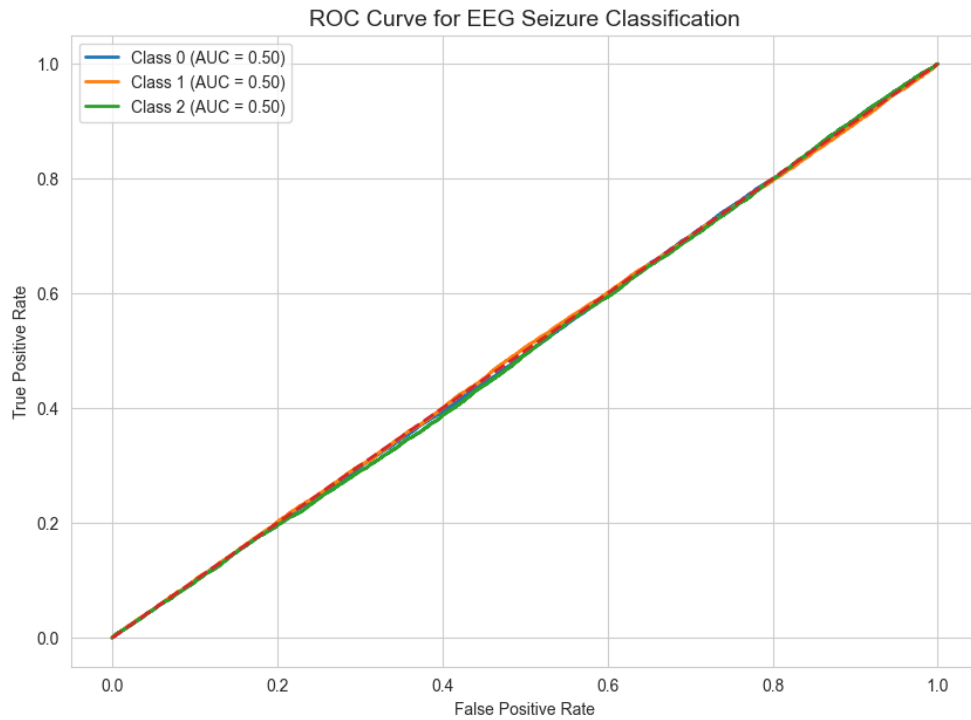


Figure 3: ROC Curve Analysis

The ROC curves closely followed the diagonal reference line for several seizure classes, indicating weak discrimination capability.

11 Precision-Recall Curve Analysis

Precision-Recall analysis was performed to evaluate seizure detection performance under imbalanced conditions.

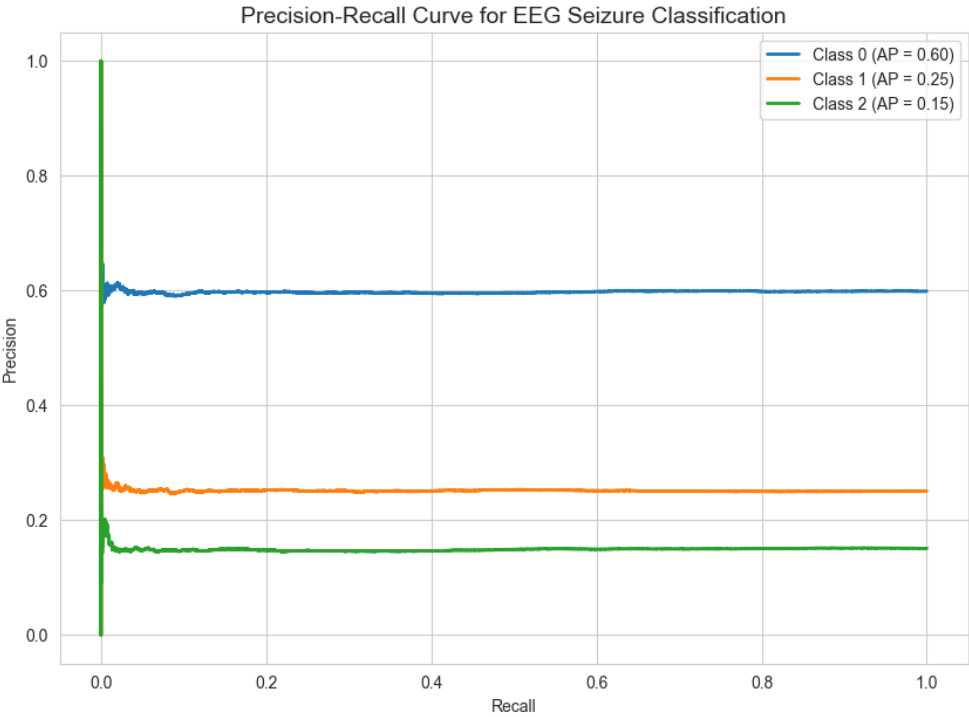


Figure 4: Precision-Recall Curve

The analysis showed:

- Higher Average Precision for majority classes
- Poor minority class recall
- Significant class prediction imbalance

These findings further confirmed the challenges of EEG seizure classification.

12 Final Comparative Analysis

Table 2: Comparative Summary of Models

Model	Performance	Observations
Baseline Logistic Regression	Moderate Accuracy	Strong majority-class bias
SMOTE + Logistic Regression	Improved Minority Recall	Better balanced learning
L1 Regularization	Sparse Features	Reduced feature complexity
L2 Regularization	Stable Learning	Limited performance improvement
Elastic Net	Balanced Regularization	Moderate generalization capability

13 Conclusion

This project presented a comparative machine learning study for EEG-based epileptic seizure prediction using Logistic Regression and multiple preprocessing strategies.

The experiments demonstrated that severe class imbalance significantly affects seizure prediction systems. Baseline Logistic Regression achieved moderate accuracy but failed to classify minority seizure classes effectively.

After applying SMOTE, minority seizure prediction improved considerably, resulting in more balanced classification performance.

Overall, the study highlights the importance of preprocessing, balanced learning, and regularization techniques in biomedical machine learning systems.